

Sandesh S. Kalantre

Experimental physicist at the intersection of
quantum hardware and AI-driven science

Knight-Hennessy Fellow, Stanford University

sandykal@stanford.edu
[Website](#) · [Google Scholar](#)

EDUCATION

Stanford University, GPA: 4.0/4.0 *June 2022 - Sept. 2026 (exp.)*
PhD in Physics, Knight-Hennessy Fellow, Advisor: Prof. David Goldhaber-Gordon

University of Maryland (UMD), College Park, GPA: 4.0/4.0 *Aug 2018 - May 2022*
Lanczos Graduate Fellow (*transferred to Stanford*) Advisor: Prof. James R. Williams

Indian Institute of Technology (IIT), Bombay - India, GPA: 9.87/10.0 *Aug 2014- May 2018*
B.Tech in Engineering Physics with Honors and Minor in CS, Institute Silver Medal

RESEARCH SKILLS

Quantum Device Fabrication

- ❖ Magnetron sputtering
- ❖ E-beam evaporation
- ❖ ALD
- ❖ Photolithography
- ❖ E-beam lithography
- ❖ Reactive ion etching
- ❖ AFM imaging
- ❖ SEM imaging

Cryogenic & Microwave Instrumentation

- ❖ Dilution refrigerators
- ❖ VTI
- ❖ VNA
- ❖ AWG
- ❖ RF reflectometry
- ❖ cQED
- ❖ High-speed digitizers
- ❖ Lock-in amplifiers

Simulation & Computational Tools

- ❖ NumPy
- ❖ SciPy
- ❖ TensorFlow
- ❖ gdspy / gdstk
- ❖ Sonnet
- ❖ HFSS
- ❖ COMSOL
- ❖ SolidWorks
- ❖ FreeCAD
- ❖ KiCad
- ❖ Metal machining
- ❖ 3D printing

RESEARCH HIGHLIGHTS

[Google Scholar](#): 12 publications and over 500 citations, h-index = 10, i10-index = 10

- ❖ **First ML-based autotuning of a quantum dot device.** Developed CNN classifiers and a closed-loop tuning algorithm that autonomously drove a semiconductor quantum dot to the single-electron regime in situ, establishing a new paradigm for automated quantum device control now adopted across the field. [Kalantre et. al.](#), *npj Quantum Information* (2019); US patent.
- ❖ **Microwave spectroscopy of a thin film quantum spin liquid candidate.** Built a NbTiN superconducting resonator platform with flip-chip integration to measure spin resonance in epitaxial TbInO₃, opening a route to GHz-frequency probes of frustrated magnetism in films too small for conventional ESR. [Kalantre et. al.](#), under review at *Physical Review X*.
- ❖ **cQED architecture for superfluid stiffness in 2D superconductors.** Engineered a microwave platform for extracting superfluid stiffness in van der Waals superconductors at GHz frequencies and mK temperatures, applicable to rhombohedral graphene and other emergent 2D superconductors.
- ❖ **Resolved puzzling Shapiro measurements in driven graphene Josephson junctions.** Developed an analytical framework based on non-linear oscillator dynamics that explained anomalous Shapiro step structure in graphene Josephson junctions, reconciling microwave response measurements with junction parameters. [Kalantre et. al.](#), *Physical Review Research* (2020).

PATENTS

- ❖ Ray-based Classification Framework for Machine Learning-based Tuning Techniques *2023*
J.P. Zwolak, **S.S. Kalantre**, J.M. Taylor, T.W. McJunkin, US Patent App. 18/028,154

RELEVANT RESEARCH EXPERIENCE

Geballe Lab for Advanced Materials, Stanford University

June 2022 - Present

Advisor: Prof. David Goldhaber-Gordon

- ❖ Developed a microwave resonance setup using NbTiN superconducting resonators for probing spin liquid thin films; built a flip-chip assembly for rapid spin resonance characterization of insulating thin films
- ❖ Designed and implemented an architecture drawing from cQED techniques for superfluid stiffness measurements of van der Waals (vdW) superconductors at GHz frequencies
- ❖ Magneto-transport of topological Kondo insulator SmB₆ films and magnetic topological insulator Eu-InAs (324) nanowires to probe surface states and chiral hinge modes respectively

Quantum Materials Lab (QMDLab), University of Maryland

August 2018 - June 2022

Advisor: Prof. James R. Williams

- ❖ Probed superconducting interfaces to topological materials and vdW heterostructures through quasi-DC transport and microwave Shapiro measurements
- ❖ Developed a theory to explain Shapiro measurements of graphene Josephson junctions building on results in non-linear dynamics of the driven pendulum problem

National Institute for Standards and Technology - Gaithersburg

May 2017 - July 2017

Advisors: Prof. Jacob M. Taylor, Prof. Justyna Zwolak

- ❖ Developed convolutional neural network (CNN) classifiers to characterize charge configurations of semiconductor quantum dots, achieving automated state recognition across a continuous parameter space
- ❖ Built the first ML-based autonomous tuning system for quantum dots, demonstrated on a live device reaching the single-electron regime without human intervention which established a new paradigm for automated quantum device control

HONORS AND AWARDS

- ❖ Knight-Hennessy Fellowship (\$60,000/yr) 2023-2026
For visionary & collaborative leaders addressing complex real-world challenges at Stanford
- ❖ Lanczos Graduate Fellowship 2018-2020
Awarded to top incoming 2-3 students (\$36,000/yr) at Joint Quantum Institute, UMD
- ❖ Institute Silver Medal for graduating at the top of Engineering Physics program, IIT Bombay 2018
- ❖ S. N. Bose Scholarship (\$2000) awarded to top 50 students from India in STEM 2017
- ❖ Institute Academic Award and R.P. Singh Memorial Award, IIT Bombay 2015-2017
- ❖ ISTernship at IST Austria (\$2500) awarded to around 40 students worldwide 2016
- ❖ 2 Gold and 1 Silver medals at International Olympiads on Astronomy and Astrophysics 2011-2013

TEACHING AND LEADERSHIP EXPERIENCE

Teaching Assistant, Stanford University

2023-2025

- ❖ TA for undergraduate E&M, quantum mechanics II; mentored student teams in the design of a nuclear quadrupole resonance setup & cryogenic probes for advanced undergraduate physics lab

Teaching Assistant, IIT Bombay

2015-2018

- ❖ TA for a range of physics & mathematics undergraduate courses on electricity and magnetism, introductory quantum physics and partial-differential equations; on two occasions was also the head TA responsible for course logistics and personnel management